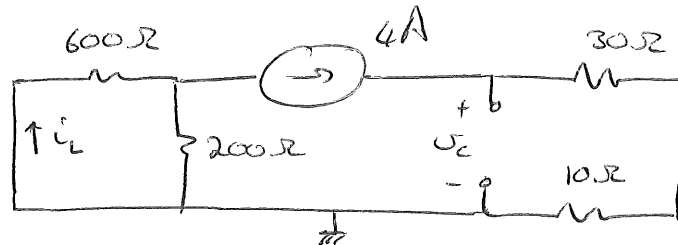
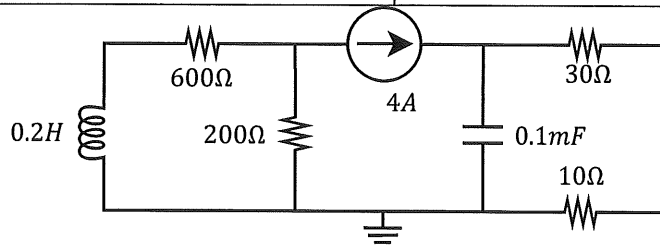


Vraag 1 [2]

Bepaal die energie gestoor in die kapasitor en induktor.

Question 1 [2]

Determine the energy stored in the capacitor and inductor.



$$* \quad i_L = \frac{4 \times 200}{600 + 200} = 1 \text{ A} \quad (\text{I-division})$$

$$w_L = \frac{1}{2} (0.2) 1^2 = 0.1 \text{ J} \quad \checkmark$$

$$* \quad U_C = 4 \times (30 + 10) = 160 \text{ V}$$

$$w_C = \frac{1}{2} (0.1 \text{ m}) (160)^2 = 1.28 \text{ J} \quad \checkmark$$

Vraag 2 [4]

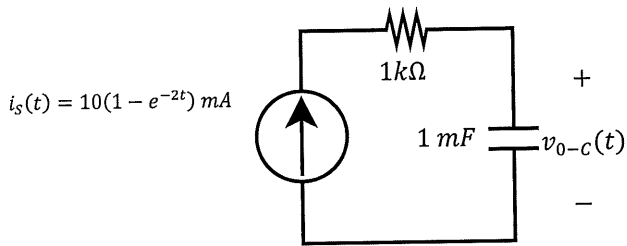
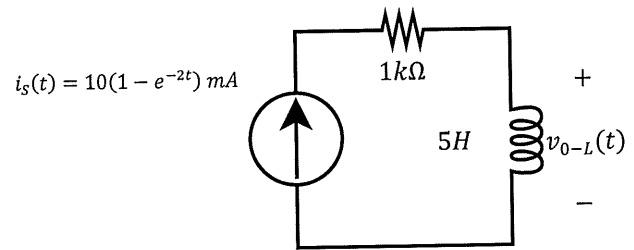
Beskou kringbane 1 en 2. In altwee gevalle word die uitsetspanning gemeet oor die reaktiewe komponent in die kringbaan. Bepaal telkens $v_{0-C}(t)$ en $v_{0-L}(t)$.

Aanvaar $v_{0-C}(0) = 1 \text{ V}$ en $v_{0-L}(0) = 0.1 \text{ V}$.

Question 2 [4]

Consider circuits 1 and 2. For both circuits the output voltage is measured across the reactive component in the circuit. Determine $v_{0-C}(t)$ and $v_{0-L}(t)$.

Assume $v_{0-C}(0) = 1 \text{ V}$ and $v_{0-L}(0) = 0.1 \text{ V}$.

Kringbaan / Circuit 1**Kringbaan / Circuit 2**

$$\begin{aligned}
 \textcircled{1} \quad v_{0-C} &= \frac{1}{C} \int_0^t i_s(t) dt + v_{0-C}(0) \\
 &= \frac{1}{1 \text{ mF}} \int_0^t 10 \times 10^{-3} (1 - e^{-2t}) dt + 1 \\
 &= 10 \int_0^t (1 - e^{-2t}) dt + 1 \\
 &= 10 \left[t + \frac{1}{2} e^{-2t} \right]_0^t + 1 \\
 &= 10t + 5e^{-2t} - 10 \left(0 + \frac{1}{2} e^{-0} \right) + 1 \\
 &= 10t + 5e^{-2t} - 5 + 1 \\
 &= \underline{10t + 5e^{-2t} - 4 \text{ V}} \quad \checkmark
 \end{aligned}$$

$$\begin{aligned}
 \textcircled{2} \quad v_{0-L} &= L \frac{di_s}{dt} \\
 &= 5 \times 10 \frac{d}{dt} (1 - e^{-2t}) \times 10^{-3} \\
 &= 50 (2e^{-2t}) \times 10^{-3} \\
 &= \underline{0.1 e^{-2t} \text{ V}} \quad \checkmark
 \end{aligned}$$

Vraag 3 [2]

Bepaal en skryf die antwoord in polêre vorm:

$$z = \frac{\sqrt{2}e^{j45^\circ}}{\sqrt{\frac{(8+8j)^*}{\sqrt{2}}}}$$

Question 3 [2]

Determine and write the answer in polar form:

$$z = \frac{\sqrt{2}e^{j45^\circ}}{\sqrt{\frac{(8+8j)^*}{\sqrt{2}}}}$$

$$Z = \frac{52 \angle 45^\circ}{\left(\frac{8-8j}{\sqrt{2}}\right)^{\frac{1}{2}}} = \frac{52 \angle 45^\circ}{\left(\frac{8\sqrt{2} \angle -45^\circ}{\sqrt{2}}\right)^{\frac{1}{2}}} = \frac{52 \angle 45^\circ}{(8 \angle -45^\circ)^{\frac{1}{2}}}$$

$$= \frac{52 \angle 45^\circ}{\sqrt{8} \angle -45/2} = \frac{52 \angle 45^\circ}{2\sqrt{2} \angle -22.5} = 0.5 \angle 67.5^\circ \checkmark$$

Vraag 4 [4]

- a) Watter van die volgende seine loop voor en met hoeveel grade:

$$\bar{V}_1 = 5 \angle 170^\circ, v_2(t) = -10 \sin(\omega t + 100^\circ)$$

- b) Vereenvoudig $i(t)$ en skryf as enkele sinusoid

$$i(t) = I_m \cos(\omega t - \theta^\circ) \text{ met } I_m > 0 \text{ en } -180^\circ \leq \theta \leq 180^\circ.$$

$$i(t) = -3 \cos(\omega t + 120^\circ) + 5 \sin(\omega t - 60^\circ) \text{ A}$$

Question 4 [4]

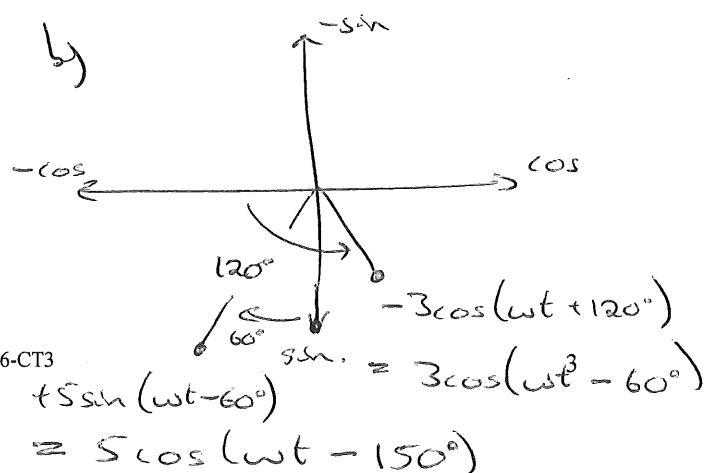
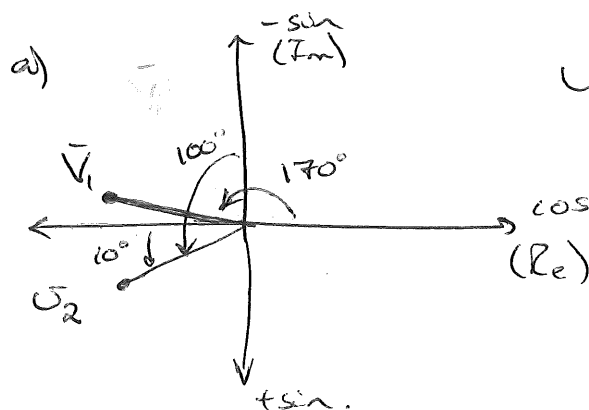
- a) Which of the following signals is leading and by how many degrees:

$$\bar{V}_1 = 5 \angle 170^\circ, v_2(t) = -10 \sin(\omega t + 100^\circ)$$

- b) Simplify $i(t)$ and write as a single sinusoid

$$i(t) = I_m \cos(\omega t - \theta^\circ) \text{ with } I_m > 0 \text{ and } -180^\circ \leq \theta \leq 180^\circ.$$

$$i(t) = -3 \cos(\omega t + 120^\circ) + 5 \sin(\omega t - 60^\circ) \text{ A}$$



$$\therefore \bar{I} = 3 \angle -60^\circ + 5 \angle -150^\circ$$

$$= 5.83 \angle -119^\circ$$

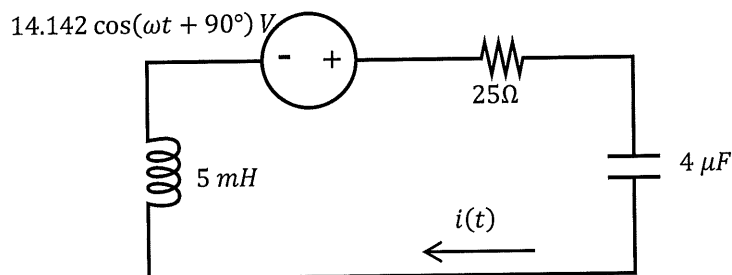
$$\therefore i(t) = 5.83 \cos(\omega t - 119^\circ)$$

Vraag 5 [2]

Bereken die fasor voorstelling van $i(t)$ en skryf die antwoord in polêre vorm. ($\omega = 5 \text{ krad/s}$)
($r \angle \theta$ met $r > 0$ en $-180^\circ \leq \theta \leq 180^\circ$)

Question 5 [2]

Determine the phasor representation of $i(t)$ and write the answer in polar form. ($\omega = 5 \text{ krad/s}$)
($r \angle \theta$ where $r > 0$ and $-180^\circ \leq \theta \leq 180^\circ$)



$$Z_L = j(5 \text{ k})(5 \text{ m}) = j25 \Omega \quad Z_C = \frac{1}{j(5 \text{ k})(4 \mu)} = -j50 \Omega$$

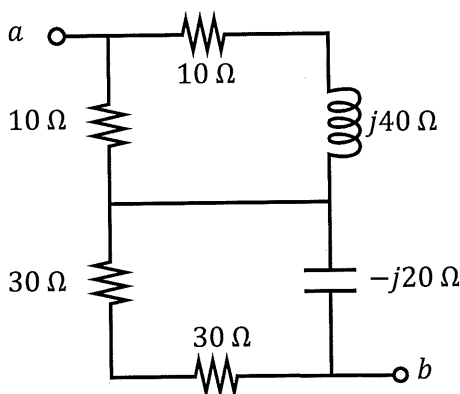
$$I = \frac{V_s}{Z_T} = \frac{14.142 \angle 90^\circ}{25 + j25 - j50} = \frac{14.142 \angle 90^\circ}{25 - j25} = 0.4 \angle 135^\circ \text{ A}$$

Vraag 6 [2]

Bepaal die ekwivalente impedansie tussen terminale a en b vir onderstaande kringbaan.
(Skryf die antwoord in **reghoekige formaat**.)

Question 6 [2]

Determine the equivalent impedance between terminals a and b for the circuit below.
(Write the answer in **rectangular format**.)



$$Z_{ab} = 10 \parallel (10 + j40) + (30 + 30) \parallel -j20$$

$$= \frac{10(10 + j40)}{20 + j40} + \frac{60(-j20)}{60 - j20}$$

$$= 9 + 2j + 6 - 18j = 15 - 16j \Omega$$